

Deep Generative Models: Diffusion Models – SDE perspective

Fall Semester 2025

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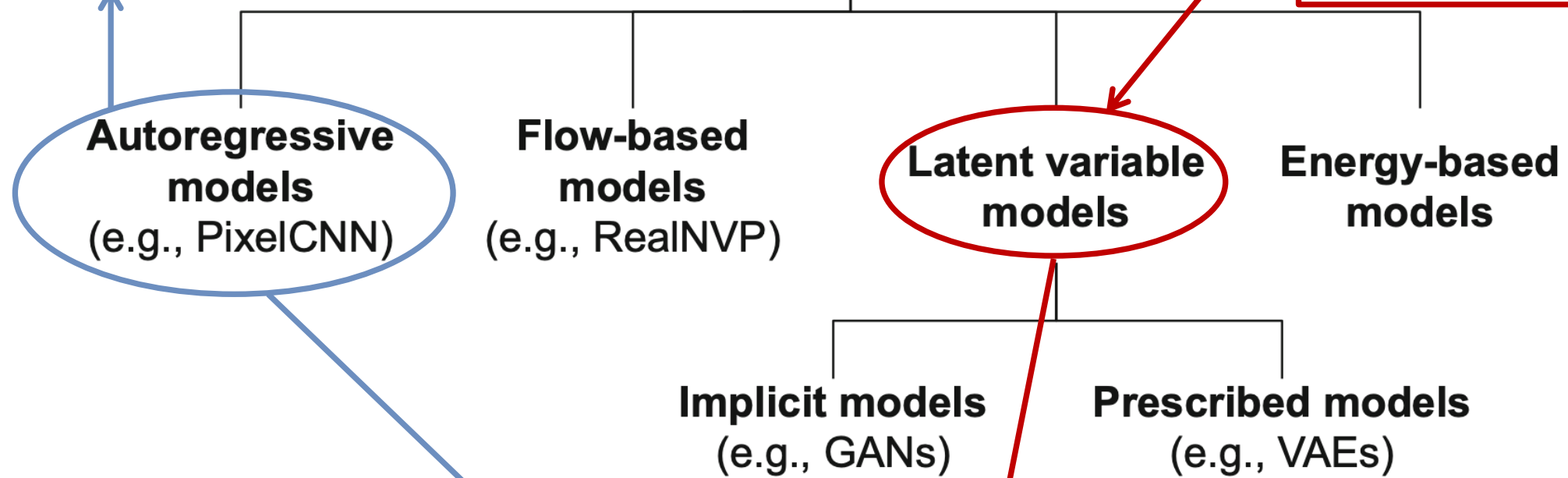


Taxonomy of Generative Models

What we've learned:
• MMs, HMMs, LDSs, RNNs, SSMs

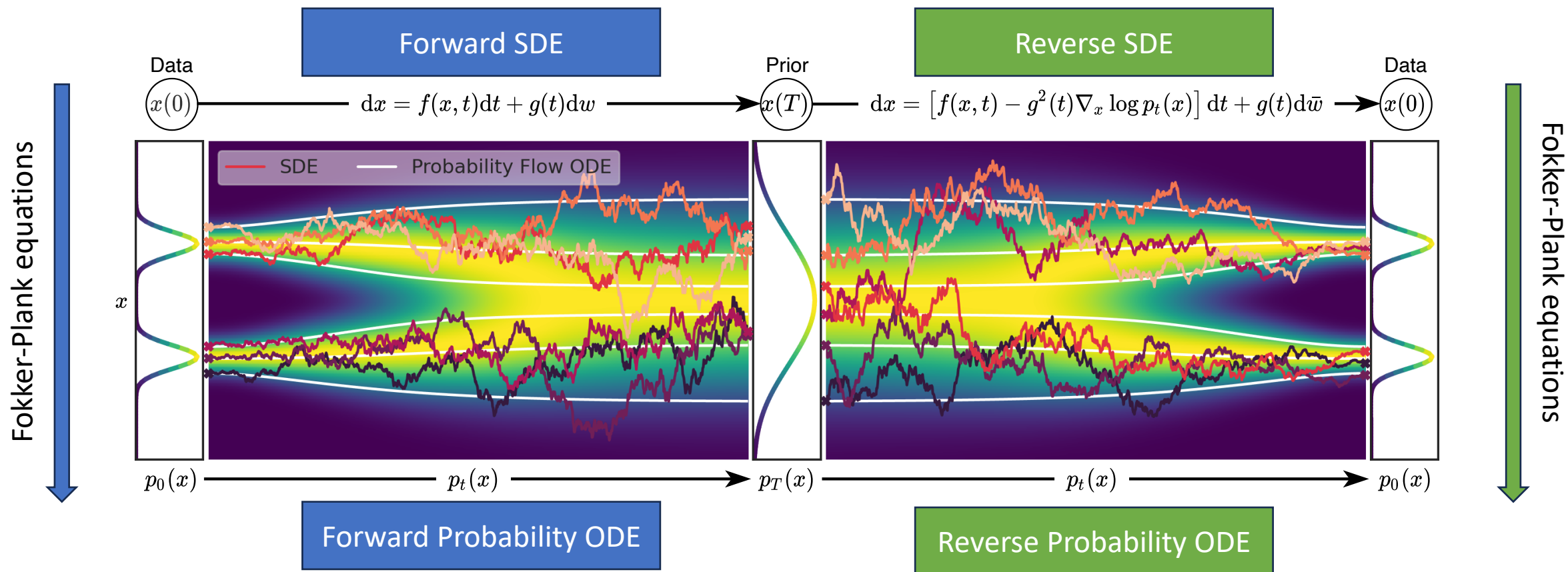
What we've learned:
• PPCA
• VAE
• Latent Diffusion

Deep Generative Models



What we study now:
• SDE perspective of Diffusion process

Stochastic Differential Equation (SDE) view



Stochastic Differential Equation

- SDE is a differential equation in which one or more terms are stochastic processes.

$$dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dw_t$$

- $\mu(x_t, t)$ is called as the **drift coefficient** and $\sigma(x_t, t)$ is called the **diffusion coefficient**.
- w_t is called the Wiener process and has the following properties:
 1. Initial zero condition: $w_0 = 0$ almost surely
 2. Independent increments: For every $t, u \geq 0$, the increments $w_{t+u} - w_t$ are independent of $w_{<t}$.
 3. Gaussian increments: For every $t, u \geq 0$, the increments $w_{t+u} - w_t \sim \mathcal{N}(0, uI)$.
 4. Continuity paths: w_t is continuous in t almost surely
- How to obtain x_t given x_0 ?

$$x_t - x_0 = \underbrace{\int_0^t \mu(x_t, t)dt}_{\text{Lebesgue integral}} + \underbrace{\int_0^t \sigma(x_t, t)dw_t}_{\text{It\^o integral}}$$

μ, σ should be Lipschitz continuous in (x, t)
for the existence and uniqueness of SDE solution

The Fokker-Plank Equation

$$dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dw_t$$

- Every SDE has an associated ODE in probability space; which is characterized by the Fokker-Plank equation.
- The probability ODE is given by:

$$\frac{dp(x_t, t)}{dt} = -\nabla_x \cdot \left(\mu(x_t, t)p(x_t, t) - \frac{1}{2} \nabla_x (\sigma(x_t, t)^2 p(x_t, t)) \right)$$

where $\nabla_x \cdot g(x_t, t) = \sum_i \partial_{x_i} (g(x_t, t))_i$ is the divergence operation.

- Conservation law: The net change of probability at point (x_t, t) is the same as the flow due to drift and diffusion coefficient.
- **Forward process:** the process from $t = 0 \rightarrow t = 1$ with initial distribution p_0 .
- **Reverse process:** the SDE with initial distribution p_1 , final distribution p_0 and probability ODE that matches the forward SDE.
- To prove the Fokker-Plank equation we will need Itô's Lemma.

Itô's Lemma: Scalar case

As $\Delta t \rightarrow 0$, it holds $(x_{t+\Delta t} - x_t)^2 = \sigma(x_t, t)^2 (w_{t+\Delta t} - w_t)^2 + o(\Delta t^{1.5})$

$w_{t+\Delta t} - w_t \sim \mathcal{N}(0, \Delta t)$

- **Itô** is pronounced as *ee-toh*.
- Suppose $y_t = u(x_t, t)$ is another process defined on x_t , then we have

$$dy_t = \partial_t u(x_t, t)dt + \partial_x u(x_t, t)dx_t + \frac{1}{2} \sigma(x_t, t)^2 \partial_x^2 u(x_t, t)dt$$

- **Intuition:** Let us compute the finite difference $u(x_{t+\Delta t}, t + \Delta t) - u(x_t, t)$ using Taylor's series:

$$u(x_{t+\Delta t}, t + \Delta t) - u(x_t, t) = \Delta t \partial_t u(x_t, t) + (x_{t+\Delta t} - x_t) \partial_x u(x_t, t) + \frac{1}{2} (x_{t+\Delta t} - x_t)^2 \partial_x^2 u(x_t, t) + o(\Delta t^{1.5}).$$

- Now as $\Delta t \rightarrow 0$, we have

$$\begin{aligned} u(x_{t+\Delta t}, t + \Delta t) - u(x_t, t) &= \Delta t \partial_t u(x_t, t) + (x_{t+\Delta t} - x_t) \partial_x u(x_t, t) + \frac{1}{2} \sigma(x_t, t)^2 (w_{t+\Delta t} - w_t)^2 \partial_x^2 u(x_t, t) + o(\Delta t^{1.5}), \\ &= \Delta t \partial_t u(x_t, t) + (x_{t+\Delta t} - x_t) \partial_x u(x_t, t) + \frac{1}{2} \Delta t \partial_x^2 u(x_t, t) + o(\Delta t^{1.5}), \\ &= \Delta t \partial_t u(x_t, t) + (x_{t+\Delta t} - x_t) \partial_x u(x_t, t) + \frac{1}{2} \Delta t \partial_x^2 u(x_t, t). \end{aligned}$$

- The actual proof requires more rigorous arguments refer to the book:
[Stochastic Calculus For Finance, by Steven Shreve.](#)

Proof of Fokker-Plank Equation: Scalar Case

$$dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dw_t$$

- From Itô's Lemma, we have that for any u it holds

$$du(x_t, t) = \partial_t u(x_t, t)dt + \partial_x u(x_t, t)dx_t + \frac{1}{2} \sigma(x_t, t)^2 \partial_x^2 u(x_t, t)dt.$$

- Using the forward SDE

$$du(x_t, t) = \left[\partial_t u(x_t, t) + \mu(x_t, t) \partial_x u(x_t, t) + \frac{1}{2} \partial_x^2 u(x_t, t) \sigma(x_t, t)^2 \right] dt + \partial_x u(x_t, t) \sigma(x_t, t) dw_t.$$

- Now we apply expectation on both sides:

$$\mathbb{E}[du(x_t, t)] = \mathbb{E} \left[\partial_t u(x_t, t) + \mu(x_t, t) \partial_x u(x_t, t) + \frac{1}{2} \partial_x^2 u(x_t, t) \sigma(x_t, t)^2 \right] dt + \mathbb{E}[\partial_x u(x_t, t) \sigma(x_t, t) dw_t].$$

$$d\mathbb{E}[u(x_t, t)] = \mathbb{E} \left[\partial_t u(x_t, t) + \mu(x_t, t) \partial_x u(x_t, t) + \frac{1}{2} \partial_x^2 u(x_t, t) \sigma(x_t, t)^2 \right] dt + \mathbb{E}[\partial_x u(x_t, t) \sigma(x_t, t)] d\mathbb{E}[w_t].$$

- We expand the expectation as integral and transfer the

Proof of Fokker-Plank Equation: Scalar Case

- We now evaluate the derivative:

$$\begin{aligned} & \int \cancel{\partial_t u(x, t)} p(x, t) dx + \int \partial_t p(x, t) \cdot u(x, t) dx \\ &= \int \left[\cancel{\partial_t u(x, t)} + \mu(x, t) \partial_x u(x, t) + \frac{1}{2} \sigma(x, t)^2 \partial_x^2 u(x, t) \right] p(x, t) dx. \end{aligned}$$

- Recall by integration by parts **[How ?]**:

$$\begin{aligned} \int \partial_t p(x, t) \cdot u(x, t) dx &= \int \left[-\partial_x (\mu(x, t) p(x, t)) + \frac{1}{2} \partial_x^2 (\sigma(x, t)^2 p(x, t)) \right] \cdot u(x, t) dx, \\ &\equiv \int \left[\partial_t p(x, t) - \partial_x (\mu(x, t) p(x, t)) + \frac{1}{2} \partial_x^2 (\sigma(x, t)^2 p(x, t)) \right] \cdot u(x, t) dx = 0. \end{aligned}$$

- As u is arbitrary choice, the integrand must be zero. Therefore,

$$\partial_t p(x, t) - \partial_x (\mu(x, t) p(x, t)) + \frac{1}{2} \partial_x^2 (\sigma(x, t)^2 p(x, t)) = 0.$$

Reversal of the diffusion process

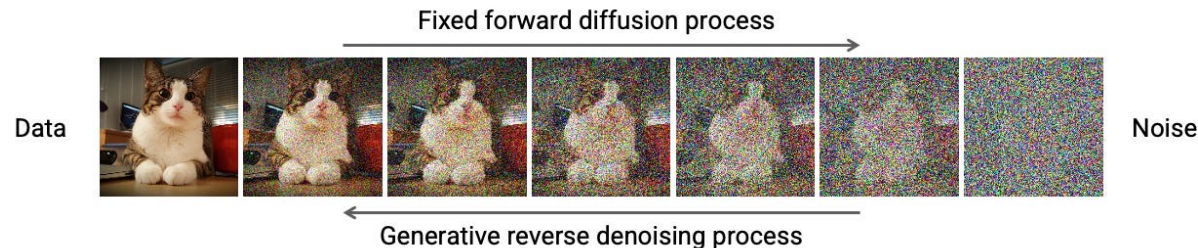
- [Brian Anderson](#) showed this in 1982 which later became a building block in [score-matching diffusion models](#) introduced in 2020.
- The forward SDE

$$dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dw_t, \quad x_0 \sim p_0$$

has the time-reversal equation:

$$dx'_t = \left(\frac{\nabla_x(p(x'_{1-t}, 1-t) \cdot \sigma(x'_{1-t}, 1-t)^2)}{p(x'_{1-t}, 1-t)} - \mu(x'_{1-t}, 1-t) \right) dt + \sigma(x'_{1-t}, 1-t)d\bar{w}_t$$

with $x'_0 \sim p_1$, and $(w_t, \bar{w}_t)$ being independent Wiener processes.



Proof of time-reversal diffusion process

$$dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dw_t$$

- The Fokker-Plank equation for the reverse SDE is given by: $\frac{dp(x'_{1-t}, 1-t)}{dt} =$

$$-\nabla_x \cdot \left(\left(-\mu(x'_{1-t}, 1-t) + \frac{\nabla_x (\sigma(x'_{1-t}, 1-t)^2 \cdot p(x'_{1-t}, 1-t))}{p(x'_{1-t}, 1-t)} \right) \cdot p(x'_{1-t}, 1-t) - \frac{1}{2} \nabla_x (p(x'_{1-t}, 1-t) \cdot \sigma(x'_{1-t}, 1-t)^2) \right),$$

$$= - \left(-\nabla_x (\mu(x'_{1-t}, 1-t) \cdot p(x'_{1-t}, 1-t)) + \nabla_x \cdot (p(x'_{1-t}, 1-t) \cdot \nabla_x \sigma(x'_{1-t}, 1-t)^2) \right)$$

$$= \frac{dp(x_{1-t}, 1-t)}{d(1-t)}.$$
- The probability ODEs match in both the direction match, i.e., reversal of the time derivative $\frac{dp(x_t, t)}{dt}$ is the same $\frac{dp(x_{1-t}, 1-t)}{d(1-t)} = \frac{dp(x'_{1-t}, 1-t)}{dt}.$
- Since the boundary conditions and ODE are the same the solutions must match.

Special case: SDE $\longrightarrow$ DDPM

- Set $\mu(x, t) = -\frac{1}{2}\sqrt{1 - \alpha_t} \cdot x_t$, and $\sigma(x, t) = N\sqrt{1 - \alpha_t}$.
- We have the forward process: $dx_t = -\frac{1}{2}\sqrt{1 - \alpha_t} \cdot x_t dt + N\sqrt{1 - \alpha_t} \cdot dw_t$.
- Now we discretize the time with N scales: $t = \frac{i}{N}$, and $\Delta t = \frac{1}{N}$.
- We can write $dx_t \approx x_{i+1} - x_i$, $-\frac{1}{2}\sqrt{1 - \alpha_t} \cdot x_t dt = -\frac{1}{2}\sqrt{1 - \alpha_i} \cdot x_i \Delta t$, and $N\sqrt{1 - \alpha_t}dw_t \sim \mathcal{N}(\epsilon_i; 0, \sqrt{1 - \alpha_t}I)$.
- We obtain $x_{i+1} = \left(1 - \frac{1}{2}\sqrt{1 - \alpha_i}\right)x_i + \sqrt{1 - \alpha_t} \cdot \epsilon_i$.
- Let $N \rightarrow \infty$, then we have $\left(1 - \frac{1}{2}\sqrt{1 - \alpha_i}\right) \approx \sqrt{1 - 1 + \alpha_i} = \sqrt{\alpha_i}$.
- Finally, we have $x_{i+1} \approx \sqrt{\alpha_i}x_i + \sqrt{1 - \alpha_t}\epsilon_i$ This is the same as DDPM.

Special case: DDPM $\longrightarrow$ SDE

- Recall the forward process in diffusion models:

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(\epsilon_t; 0, I)$$

This is in the discrete time.

- Define $\alpha(t) = N \cdot (1 - \alpha_t)$, $\Delta t = \frac{1}{N}$, where N is the number of noise scales.

- In the limit $N \rightarrow \infty$ we have,

- $x_{t+\Delta t} = \sqrt{1 - \alpha(t)\Delta t} \cdot x_t + \sqrt{\Delta t \cdot \alpha(t)} \cdot \epsilon_t,$
- $x_{t+\Delta t} \approx \left(1 - \frac{1}{2}\right) x_t + \sqrt{\alpha(t)} \cdot \sqrt{\Delta t} \cdot \epsilon_t,$
- $x_{t+\Delta t} - x_t \approx -\frac{\alpha(t)}{2} x_t \cdot \Delta t + \sqrt{\alpha(t)} \cdot \sqrt{\Delta t} \cdot \epsilon_t,$
- $dx_t \approx -\frac{\alpha(t)}{2} x_t dt + \sqrt{\alpha(t)} \sqrt{dt} \cdot \epsilon_t.$

- $\sqrt{dt} \cdot \epsilon_t$ is infinitesimal Normal random variable or Brownian / Wiener process.

Special case: SDE $\longrightarrow$ Score-based models

- Set $\mu(x, t) = 0$, and $\sigma(x, t) = \sigma(t)$.

Forward: $dx_t = \sigma(t)dw_t; x_0 \sim p_0$,

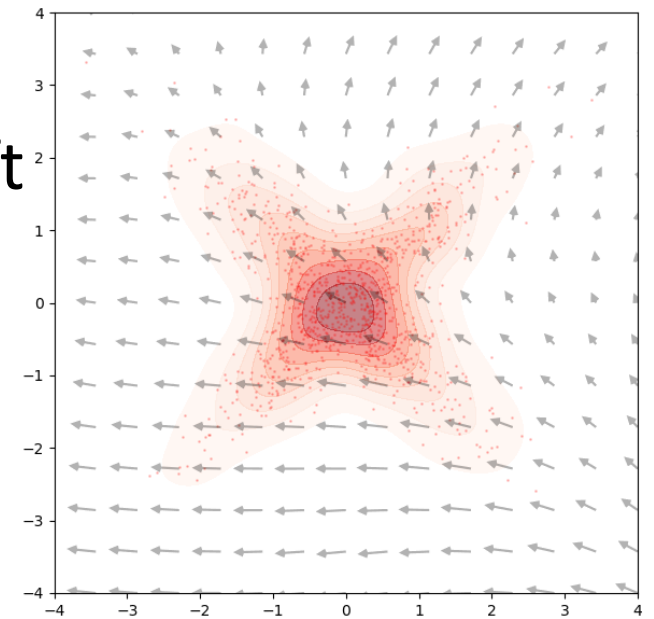
Reverse: $dx'_t = \sigma(1 - t)^2 \underbrace{\nabla_x \ln(p(x'_{1-t}, 1 - t))}_{\text{Stein's score function}} dt + \sigma(1 - t)d\bar{w}_t; x'_0 \sim p_1.$

Stein's score function

- In the reverse process, we are forcing the samples to drift to high-likelihood regions just like in score-based diffusion model.

- After discretizing the reverse process, we obtain:

$$x'_{t+\Delta t} = x'_t + \sigma(1 - t)^2 \Delta t \nabla_x \ln(p(x'_{1-t}, 1 - t)) + \sigma(1 - t) \sqrt{\Delta t} \epsilon_t$$



Vector field of score function